

When Doctors Start Prescribing Electronics, What Would the Dosage Be



Janani Gopalakrishnan
Vikram is a technically-qualified freelance writer, editor and hands-on mom based in Chennai

The way the term medical electronics is perceived by us has constantly evolved over time. At one time, it meant digital instruments to read one's blood pressure or glucose levels, or large, Internet-connected meters that could be used by doctors to test and treat patients in remote locations (with the help of trained technical staff) or implantable devices like cardiac pacemakers. However, today the term has taken on whole new definitions, which are unimaginable and, even, unquantifiable.

From wearable devices that work with your mobile phone to monitor and convey your health condition to a physician, and devices that make sure you have your medicines and electronics built into your house's walls to monitor your heart rate, to implantable devices that help your body work better, modern medical electronics is at the cusp of a new era of medical diagnosis and treatment. This is fuelled by trends like flexible, organic electronics, 3D printing and the Internet of Things (IoT).

Here we look at some futuristic research

and development (R&D) in this space, which promise a lot of excitement and good health in the days to come.

Wearables and Big Data: A magical duo

Although it might seem as if the market is flooded with just fitness enhancers, some start-ups are already making some clinically-significant wearable devices, which have the potential to eventually bridge the gap between physicians and patients, making real-time medicine a reality.

Muse is a good example. It uses advanced sensor technology to fit the functioning of a clinical-grade electroencephalogram (EEG) into a beautiful and comfortable headband, which works hand-in-hand with an app to aid care givers an understanding and helping them out in conditions like attention deficit hyperactivity disorder (ADHD), anxiety and depression.

Another is Resound, an iPhone-connected hearing aid that can be customised according to location and situation.



Muse headband
(Image courtesy: www.choosemuse.com)

